# Predicting Order Cancellations in Pakistan's E-commerce Market

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# 1 Problem Identification

## 1.1 Input

The input dataset consists of transactional records from a Pakistani e-commerce platform spanning from January 2016 to August 2018. This dataset was published by Zeeshan Usmani on Kaggle (Usmani, 2018).

Each record represents a single order and includes features such as item price, quantity ordered, product category, discount amount, order date, customer ID, customer registration date, and payment method. The dataset was filtered to include Cash-on-Delivery (COD) orders only, as this payment method dominates Pakistan's e-commerce landscape and is associated with higher cancellation risk.

## 1.2 Output

The output is a binary classification label — *Successful* or *Cancelled* — indicating whether a given COD order was successful or cancelled.

## 1.3 Objective

The objective of this project is to build and compare various supervised machine learning models that can predict the likelihood of a COD order being cancelled based on its transactional features.

By identifying the most accurate and reliable model, the study aims to provide e-commerce platforms with a tool for early detection of high-risk orders, enabling proactive interventions such as order verification, delivery prioritisation, or targeted customer engagement to reduce cancellation rates and associated revenue losses.

# 2 Background Summary

Pakistan's e-commerce market has grown rapidly over the past decade, driven by rising smartphone usage, a young digitally-connected population, and increased investment in digital infrastructure. The market generated around USD 5.8 billion in revenue in 2025, with 10–15% growth expected in 2026 (ECDB, 2025).

However, this growth comes with a major challenge: high order cancellation rates. While global cancellation rates typically range between 16–17%, markets that rely heavily on Cash on Delivery (COD) experience significantly higher rates (McKinsey & Company, 2022).

This creates a serious operational problem. A large portion of orders result in wasted effort — including packaging, shipping, and returns — without generating revenue. These cancellations also disrupt inventory planning, increase customer acquisition costs, and tie up working capital.

COD is a key driver of cancellations in Pakistan. Since customers pay only at the time of delivery, they face no commitment upfront. As a result, many orders are cancelled at the doorstep, which is the most expensive stage in the delivery process.

In the event of order cancellation, the order must be returned to the seller, thereby doubling logistics costs. To make matters worse, additional expenses are also coupled on to the existing cost in the form of handling, quality checks, and restocking.

This behaviour is largely driven by a lack of trust in online transactions, especially among first-time buyers (World Bank, 2022). Customers often use COD as a safety mechanism, declining orders if expectations have not been met. Until trust and digital payment adoption culture improve, COD will continue to dominate the market and contribute to higher cancellation rates (State Bank of Pakistan, 2023).

### 2.1 Business Impact

The financial impact of cancellations is significant. For a mid-sized business processing 10,000 monthly orders, a cancellation rate of 9.6% (taken from our dataset) would lead to 960 failed deliveries in a month and over 11,000 returned orders around the year.

For larger platforms such as Daraz and GoLootlo, even small improvements in prediction accuracy can translate into retained revenue, lower logistics costs, and better seller economics.

## 3 Existing Literature Review

Academic work on e-commerce cancellation prediction has grown; however, it still remains under-explored compared to related areas such as churn prediction. The research broadly focuses on two questions: which algorithms operate best on transactional data, and how to handle practical challenges like class imbalance and asymmetric misclassification costs.

Şahinbaş (2022) compared neural networks, SVMs, logistic regression, XGBoost, and Random Forest on Turkish e-commerce data, finding that ensemble methods — particularly XGBoost and tuned Random Forest — consistently outperformed simpler models. The Turkish context is relevant here since Turkey shares the COD structure of Pakistan and faces similar logistics pressures.

In addition, Chidroop and Moharir (2020) tested boosted decision trees, decision forests,

neural networks, and SVMs on Indonesian e-commerce data, with a useful contribution being their simulation of cold-start scenarios — running models with and without customer history. This is directly applicable since many orders in our dataset come from infrequent or first-time buyers.

### 3.1 The Gap in Literature

Although existing studies show that machine learning can effectively predict cancellations, they do not focus on Pakistan's unique context. Specifically, they do not address:

- High reliance on COD
- Doorstep cancellations
- Severe class imbalance
- Unequal cost of prediction errors

This project addresses this gap by analysing over 200,000 COD transactions from Pakistan's largest publicly available e-commerce dataset, comparing various classifiers, and translating the findings into actionable business recommendations.

## 4 Data Cleaning and Feature Engineering

The raw dataset exported from Kaggle contained 584,524 rows and 21 columns. Since the number of missing values was small in comparison to the entire dataset, these rows were dropped and the dataset was filtered to include Cash-on-Delivery (COD) orders only. Non-predictive columns such as `increment_id`, `commission_code`, `BI Status`, `MV`, `item_id`, and `payment_method` were also dropped. The transformed dataset comprises 213,291 rows.

Several features were extracted and engineered from this dataset with the aim of capturing behavioural and temporal patterns that may influence order cancellations. These include:

- `discount_pct` — discount percentage
- `has_discount` — boolean value
- `item_total` — price × qty_ordered
- `is_repeat` — whether the customer was a repeat or one-time customer
- `price_bracket` — binned price ranges
- `day_name` — day of the week when the order was placed

The following sections highlight the univariate and bivariate analysis carried out on these features.

# 5 Exploratory Data Analysis

## 5.1 Univariate Analysis

### 5.1.1 Order Status

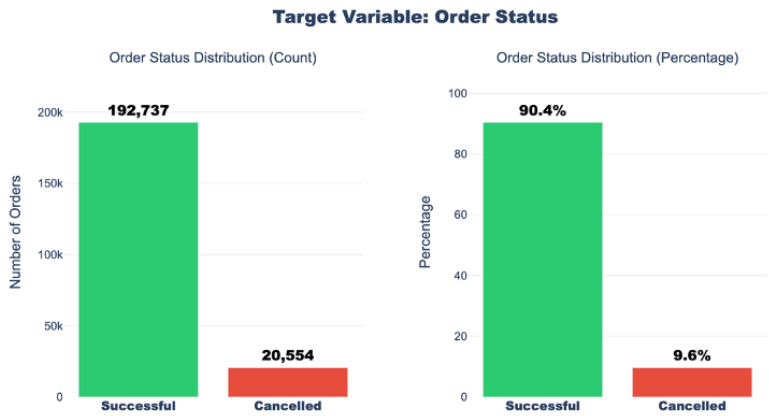


Figure 1: Distribution of order status showing class imbalance in the dataset.

Figure 1 reveals a significant class imbalance with respect to the target variable. Out of a total of 213,291 orders placed, 192,737 (90.4%) were successful whereas 20,554 (9.6%) were cancelled by the customer. The successful class outnumbers the cancelled class by roughly 9:1. This class imbalance will be addressed during the modelling phase of the project with particular emphasis on Precision, Recall, and F1-score.

### 5.1.2 Product Category

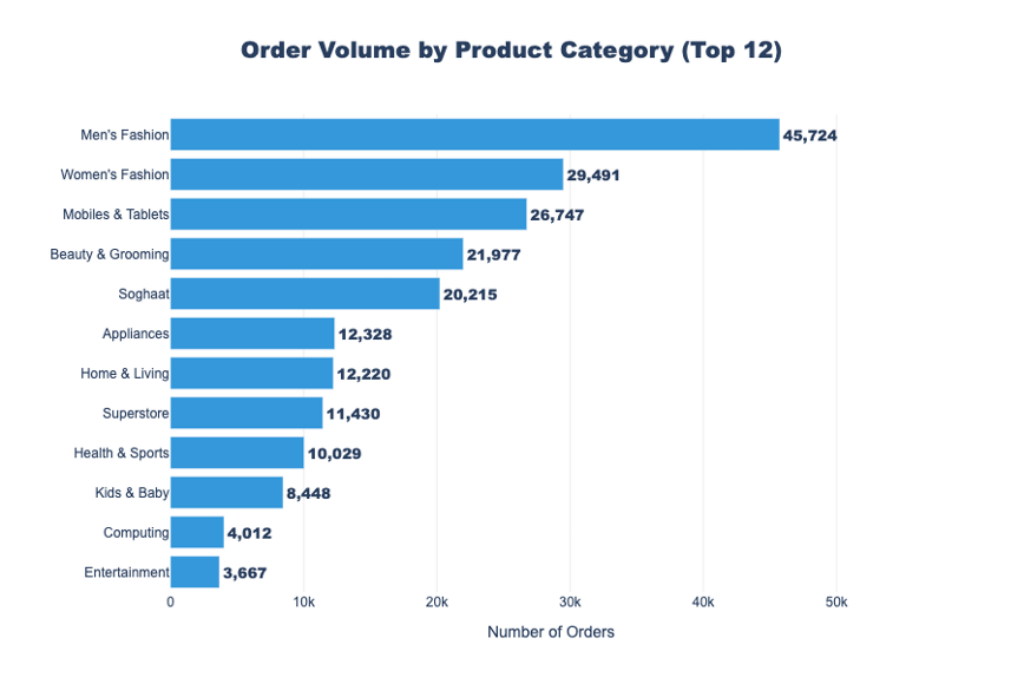


Figure 2: Bar chart showing order volume for top 12 product categories (2016–2018).

Figure 2 illustrates that the e-commerce industry in Pakistan was dominated by Men's Fashion, followed by Women's Fashion and then Mobiles & Tablets from 2016 to 2018. These top three product categories account for nearly half of all orders (∼48%), indicating greater demand for fashion and electronic goods.

The middle tier is dominated by Lifestyle and Household categories. Soghaat (a gift category for traditional and local items) ranks $5^{\text{th}}$ in total order volume, suggesting that culturally specific products are quite popular as well. In the lower tier, categories range from Health & Sports (10,029 orders) to Kids & Baby (8,448), representing niche segments with fewer orders.

### 5.1.3 Price

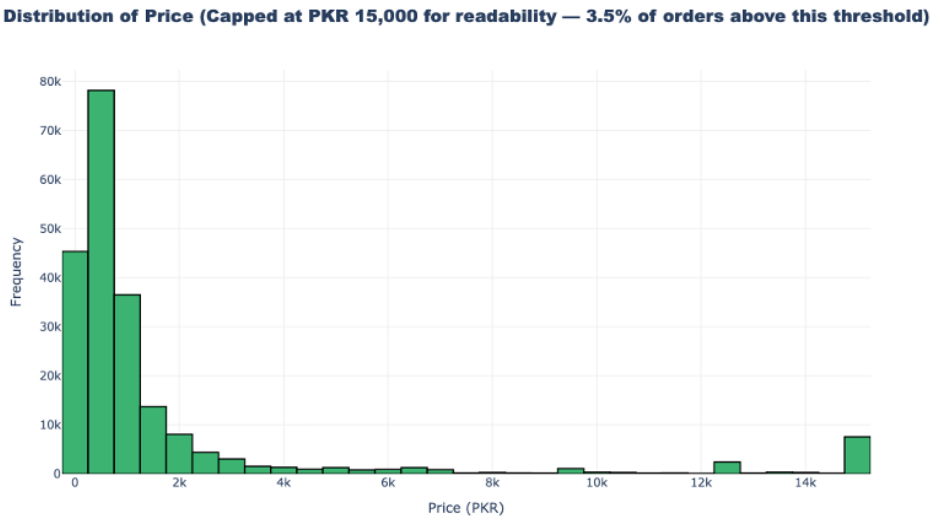


Figure 3: Histogram showing price distribution for various e-commerce orders (2016–2018).

The price distribution shown in Figure 3 is heavily right-skewed. The vast majority of orders are concentrated in the lower price range (PKR 0–2,000). This behaviour is typical for the Pakistani e-commerce market where bulk transactions comprise low-cost, everyday items such as fashion accessories and phone cases.

The long right tail with a visible peak between PKR 12,000–14,000 is likely driven by electronics and household appliances. The histogram was capped at PKR 15,000 (with 3.5% of total orders exceeding this threshold) to improve readability.

Price is likely to be an important predictor of cancellation behaviour. Low-price orders may see higher cancellation rates due to low commitment, while high-price orders may also incur cancellations due to buyer's remorse, payment failures, and longer delivery times for premium products.

### 5.1.4 Discount

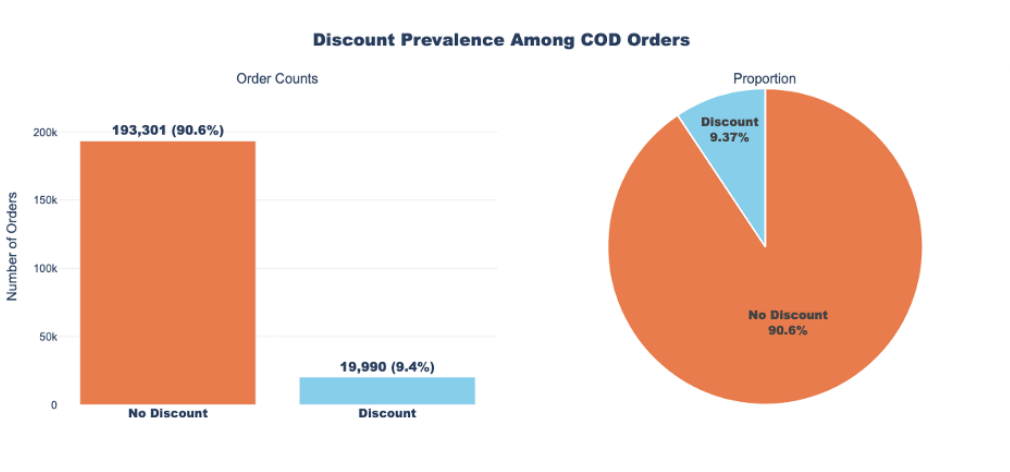


Figure 4: Bar chart (left) and pie chart (right) showing distribution of orders with respect to discount.

From Figure 4, we can infer that the large majority of COD orders (193,301) did not offer any discount. Similar to the target variable, a heavy imbalance is present.

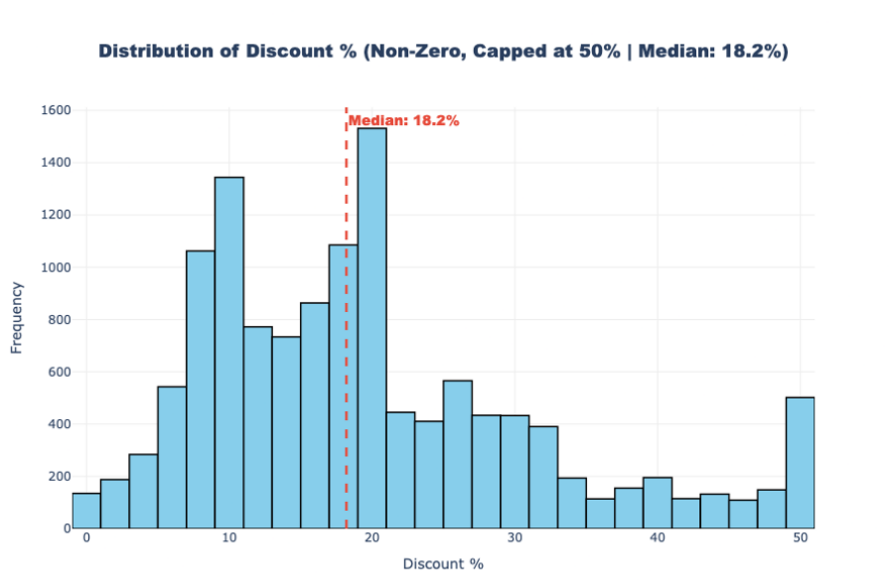


Figure 5: Histogram showing distribution of orders where discount was offered.

A closer inspection of discounted orders (9.4% of total orders) in Figure 5 reveals a peak in the 10–20% range with additional small clusters around 30% and 50%. This suggests that discounts follow structured tiers — a common practice in e-commerce marketing campaigns and flash sales.

Among the discounted orders, the median discount is 18.2% (approximately 20%). Since the discount amount appears to be recorded at the order level instead of the item level,

this leads to data quality issues and results in inflated percentages for individual line items. To mitigate this inconsistency, discounts were capped at 50%. Given this issue, the binary feature `has_discount` should provide a more reliable signal for classification; however, both features will be included in the model to determine which carries higher predictive value.

### 5.1.5 Monthly Volume

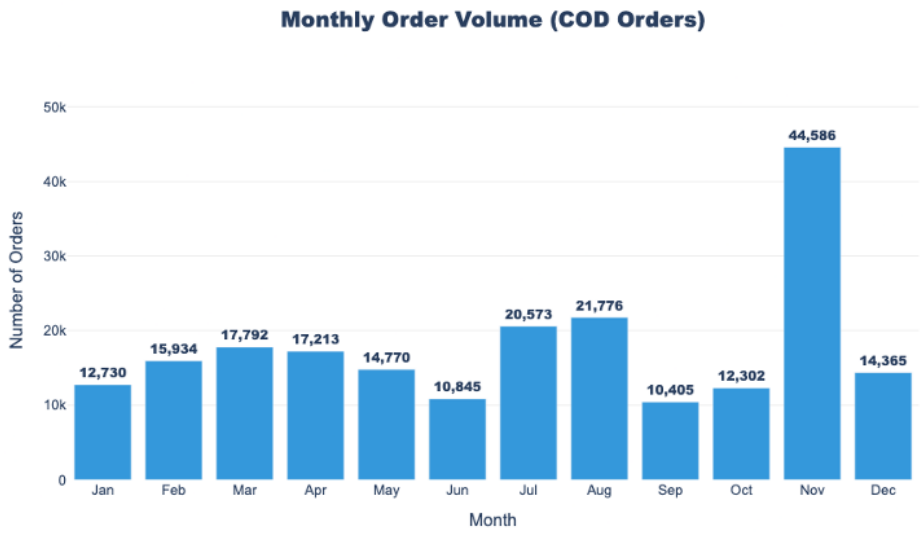


Figure 6: Bar chart showing e-commerce orders placed across different months.

Figure 6 reveals clear seasonal patterns. November is the most popular month for placing orders, coinciding with 11.11 and Black Friday Sales. Orders placed in November (44,586) are more than twice the number of orders placed in any other month, accounting for roughly 21% of total orders.

Two moderate peaks occur in March–April and July–August, potentially attributable to Pakistan Day sales and Independence Day promotions respectively. June and September appear to be the quietest months, likely falling between major promotional periods.

Seasonal patterns could influence cancellation rates: low-volume months may see more committed buyers whereas high-volume months may have higher cancellation rates due to congested delivery networks, inventory mismatches, and increased impulse buying from aggressive flash sales.

## 5.2 Bivariate Analysis

### 5.2.1 Cancellation Rate by Product Category

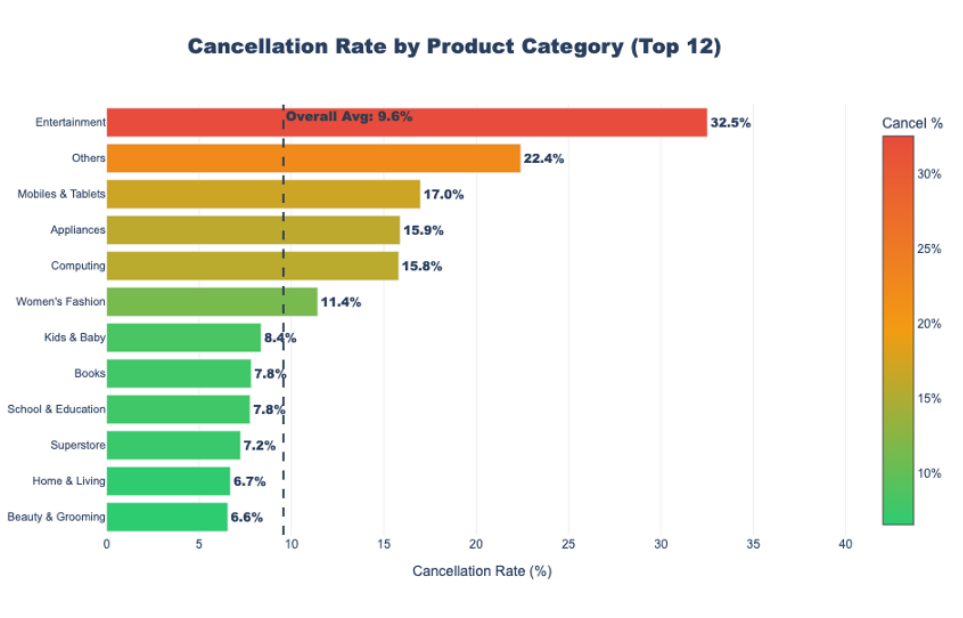


Figure 7: Bar chart showing cancellation rate by product category.

Analysis of cancellation rate by product category (Figure 7) reveals three natural tiers: **high-risk** (Entertainment, Others, Mobiles & Tablets at more than 17%), **moderate** (Appliances, Computing, Women's Fashion between 11–16%), and **low-risk** (everything from Kids & Baby down at under 9%).

Entertainment has the highest cancellation rate at 32.5% — three times the overall average of 9.6%. On the other hand, everyday essentials perform best: Beauty & Grooming (6.6%), Home & Living (6.7%), and Superstore (7.2%) have the lowest rates.

### 5.2.2 Cancellation Rate by Price Bracket

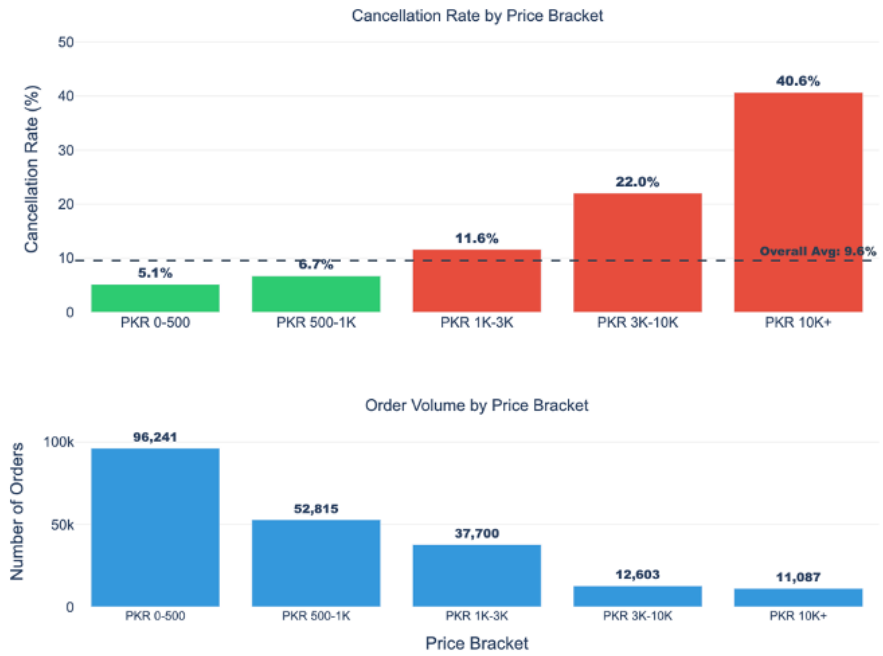


Figure 8: Cancellation rate by price bracket (top) and order volume by price bracket (bottom).

The upper bar chart in Figure 8 reveals a strong positive correlation between price and cancellation rate. The increment is almost linear from 5.1% at the lowest bracket (PKR 0–500) to 40.6% at the highest (PKR 10K+) — nearly 8 times higher.

At 40.6%, nearly half of all high-value orders get cancelled. Although this bracket has the smallest volume (11,087 orders), the revenue loss per cancellation is the highest, making it disproportionately impactful on the bottom line.

A sharp jump is observed above PKR 3K: the cancellation rate more than doubles from 11.6% (PKR 1K–3K) to 22.0% (PKR 3K–10K), suggesting PKR 3,000 as a psychological threshold where customers become significantly more hesitant.

Retention strategies such as order confirmation calls for PKR 3K+ orders, instalment payment options for PKR 10K+, and limited-time price locks could help customers commit to higher-value purchases.

### 5.2.3 Cancellation Rate by Discount

Figure 9: Bar chart showing cancellation rate for discounted vs. non-discounted orders.

Discounted orders have a cancellation rate of just 5.3%, nearly half the overall average of 9.6%, while non-discounted orders sit slightly above average at 10.0%. This confirms that discounts create a perceived "good deal" effect, strengthening the customer's commitment to follow through. The `has_discount` binary feature is therefore expected to carry meaningful predictive value.

### 5.2.4 Analysis of Remaining Features

Univariate and bivariate analysis was also carried out for month, year, day of week, customer type (one-time vs. repeat), and quantity ordered. These features revealed weak predictive signals — order volume was relatively uniform across weekdays, one-time and repeat customers showed similar cancellation rates (11.8% vs. 9.2%), and 87% of orders were single-item purchases, making quantity and `item_total` effectively equivalent to price. Detailed visualisations for these features are included in the accompanying Jupyter Notebook. The features are retained for modelling but are not presented here for brevity.

## 5.3 Correlation Heatmap

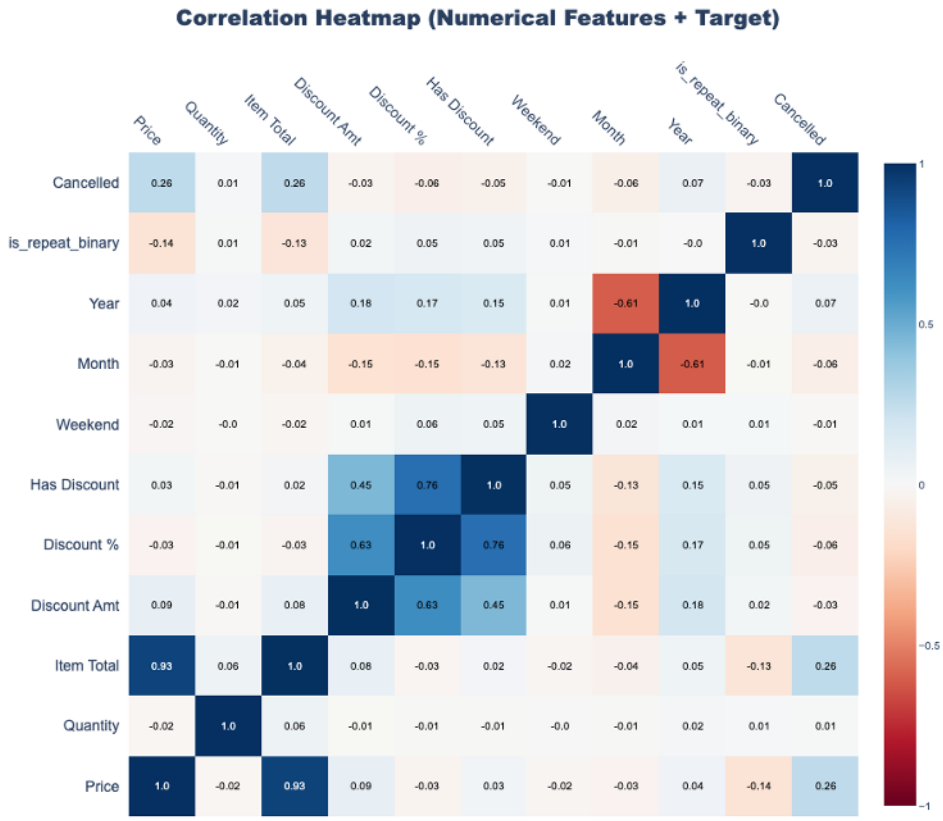


Figure 10: Correlation heatmap of numerical features.

Overall, the correlation heatmap (Figure 10) reveals weak linear correlations between individual numerical features and the target variable, with the strongest being price at only 0.26. This is consistent with expectations for tabular classification problems where cancellation behaviour is driven by non-linear interactions between multiple features rather than a single linear predictor.

The heatmap also highlights redundant features: Price and Item Total have a very high correlation of 0.93, as expected since `item_total` is derived directly from price and quantity. Similarly, the three discount features (`discount_amount`, `discount_pct`, and `has_discount`) are highly overlapping (0.45–0.76), so dimensionality reduction or feature selection can be applied to keep only one from each group during preprocessing.

Earlier bivariate analysis uncovered meaningful patterns — such as the sharp rise in cancellation rates above PKR 3,000 and significant variation across product categories. Such observations cannot be easily captured by simple correlation numbers. Models like Random Forest and Decision Trees are well-suited here, as they can capture these

complex, non-linear patterns. These will be explored in further detail in the next phase of the report.

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